

Actuarial Society of India

Examinations

May 2006

CT4 (103) - Stochastic Modelling

Indicative Solutions

Questions for CT - 4

1. The levels of discount in a car insurance no claims discount scheme are given by 0%, 10%, 20% and 40%. If the driver does not make a claim in any one year he moves up a level (e.g. from 10% to 20%); if he is already at 40% he stays there. If he makes a claim he moves down a level; if he is at 0% he stays there. A somewhat reckless driver has a probability of 0.5 of making a claim in any year. The full premium is Rs. 10,000 and the driver is currently at the 0% level. Calculate
 - i The average premium he pays per year in the long run
 - ii The expected time it will take him to reach the 40% level for the first time.
 - iii The expected value of the total premiums he will pay till he reaches 40% level for the first time.

[Marks = 4 + 3 + 3 = 10]

2. The ticket office at a metro station has a single ticket machine that is used by the passengers to buy the tickets. The machine has a tendency to break down and then has to be repaired. The time till the machine breaks down and the time required to repair the machine both follow the exponential distribution.

Let $P_{1i}(t)$, $i = 0, 1$, be the probability that at time ($t > 0$) there are i ticket machines working at the metro station, given that the ticket machine is working at time $t = 0$.

- (i.) Derive the Kolmogorov forward equations for P_{1i} , $i = 0, 1$ in terms of
 - σ , where $\frac{1}{\sigma}$ is the mean time to breakdown for a machine
 - ρ , where $\frac{1}{\rho}$ is the mean time repair a machine
- (ii.) Show that $P_{10}(t) = \frac{\sigma}{\sigma+\rho} (1 - e^{-(\sigma+\rho)t})$ deduces the value of $P_{11}(t)$
- (iii.) The station manager is considering adding an addition (identical) machine though there is only one repairman to work on both machines if both are out of action simultaneously. Assuming that the second machine is added and operates independently of the original one:
 - (a) State the generator matrix of the Markov jump process X_t which counts the number of working ticket machines at time t
 - (b) Derive the Kolmogorov forward differential equations for $p_i(t)$, $i = 0, 1, 2$, the probability that i ticket machines are working
 - (c) Given that, for some t , $p_0(t) = \frac{2\sigma^2}{2\sigma^2+2\rho\sigma+\rho^2}$, $p_1(t) = \frac{2\rho\sigma}{2\sigma^2+2\rho\sigma+\rho^2}$ and $p_2(t) = \frac{\rho^2}{2\sigma^2+2\rho\sigma+\rho^2}$ show that

$$\frac{d}{dt}p_i(t) = 0 \quad \text{for } i = 0, 1, 2$$

- (d) State what conclusions you can draw from part (c)

[Marks = 4 + 6 + 8 = 18]

3. Let the rows and columns of a finite non-negative square matrix be such that each of its row sums and each of its column sums are equal to 1. Show by using equilibrium equations that if a Markov chain has a transition matrix as defined in the earlier sentence, then the distribution that gives equal probabilities to all states is an equilibrium distribution.

[Marks = 7]

4. Suppose that $N(t)$ is a Poisson process with rate λ . Show that for $s < t$ and $n \geq k$,

$$P(N(s) = k | N(t) = n) = \binom{n}{k} p^k (1-p)^{n-k}$$

where p is to be determined.

[Marks = 8]

5. Suppose that $N(t)$ and $M(t)$ are two independent Poisson processes with rates λ and μ respectively. Show that $N(t) + M(t)$ is also a Poisson process with rate $(\lambda + \mu)$.

[Marks = 7]

Solutions

Solution to question 1:

The transition matrix is given by $\begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \end{pmatrix}$

(i.) The stationary distribution is given by:

$$(\pi_1 \pi_2 \pi_3 \pi_4) \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \end{pmatrix} = (\pi_1 \pi_2 \pi_3 \pi_4)$$

Thus,

$$\begin{aligned} \frac{\pi_1 + \pi_2}{2} = \pi_1 & \quad \frac{\pi_1 + \pi_3}{2} = \pi_2 & \quad \frac{\pi_2 + \pi_4}{2} = \pi_3 & \quad \frac{\pi_3 + \pi_4}{2} = \pi_4 \\ \Rightarrow \pi_1 = \pi_2 = \pi_3 = \pi_4 & \quad \Rightarrow \pi_1 = \pi_2 = \pi_3 = \pi_4 = \frac{1}{4} \end{aligned}$$

Thus, in the long run, the average premium is

$$\text{Rs.} \left(\frac{10000 + 9000 + 8000 + 6000}{4} \right) = \text{Rs.} 8250.$$

- (ii.) Let e_0 be the expected time to reach 40% level starting at 0% level.
Let e_1 be the expected time to reach 40% level starting at 10% level.
Let e_2 be the expected time to reach 40% level starting at 20% level.

Then

$$e_0 = 1 + \frac{e_0}{2} + \frac{e_1}{2} \tag{1}$$

$$e_1 = 1 + \frac{e_0}{2} + \frac{e_2}{2} \tag{2}$$

$$e_2 = 1 + \frac{e_1}{2} \tag{3}$$

(1) + (2) + (3) implies $e_0 + e_1 + e_2 = 3 + e_0 + e_1 + e_2 - \frac{e_2}{2}$. Thus $e_2 = 6$ and $e_1 = 10$.

Hence

$$e_0 = 1 + \frac{e_0}{2} + 5 \quad \Rightarrow e_0 = 12.$$

- (iii.) Let P_0 be the expected total premium till 40% level starting at 0% level.
Let P_1 be the expected total premium till 40% level starting at 10% level.
Let P_2 be the expected total premium till 40% level starting at 20% level.

Then

$$P_0 = 10000 + \frac{P_0}{2} + \frac{P_1}{2} \quad (4)$$

$$P_1 = 9000 + \frac{P_0}{2} + \frac{P_2}{2} \quad (5)$$

$$P_2 = 8000 + \frac{P_1}{2} \quad (6)$$

(4) + (5) + (6) implies $P_0 + P_1 + P_2 = 27000 + P_0 + P_1 + P_2 - \frac{P_2}{2}$. Thus $P_2 = 54000$ and $P_1 = 92000$.

Hence

$$P_0 = 10000 + \frac{P_0}{2} + \frac{92000}{2} \Rightarrow P_0 = 112000.$$

Solution to question 2:

(i.)

$$P_{10}(t+h) = (1-\rho h)P_{10}(t) + \sigma h P_{11}(t) \Rightarrow \frac{d}{dt}P_{10}(t) = -\rho P_{10}(t) + \sigma P_{11}(t)$$

[Equation (1)]

and

$$P_{11}(t+h) = \rho h P_{10}(t) + (1-\sigma h)P_{11}(t) \Rightarrow \frac{d}{dt}P_{11}(t) = \rho P_{10}(t) - \sigma P_{11}(t)$$

[Equation (2)]

(ii.) $P_{10}(t) + P_{11}(t) = 1$. Now following on from equation (1) of the earlier part,

$$\begin{aligned} \frac{d}{dt}P_{10}(t) + (\sigma + \rho) P_{10}(t) &= \sigma \\ \Rightarrow \frac{d}{dt}e^{(\sigma+\rho)t}P_{10}(t) &= \sigma e^{(\sigma+\rho)t} + C \\ \Rightarrow P_{10}(t) &= \frac{\sigma}{\sigma + \rho} \left(1 - e^{-(\sigma+\rho)t}\right) \text{ since } P_{10}(0) = 0. \end{aligned}$$

Therefore,

$$P_{11}(t) = 1 - \frac{\sigma}{\sigma + \rho} \left(1 - e^{-(\sigma+\rho)t}\right) = \frac{\rho + \sigma e^{-(\sigma+\rho)t}}{\sigma + \rho}$$

(iii.) (a) The generator matrix is

$$\begin{pmatrix} -\rho & \rho & 0 \\ \sigma & -(\sigma + \rho) & \rho \\ 0 & 2\sigma & -2\sigma \end{pmatrix}$$

(b) Hence, the forward equations are:

$$\begin{aligned}\frac{d}{dt}p_0(t) &= -\rho p_0(t) + \sigma p_1(t) \\ \frac{d}{dt}p_1(t) &= \rho p_0(t) - (\sigma + \rho)p_1(t) + 2\sigma p_2(t) \\ \frac{d}{dt}p_2(t) &= \rho p_1(t) - 2\sigma p_2(t)\end{aligned}$$

- (c) Simply substitute the required values from the previous part to obtain the necessary result.
- (d) The implication is that the given distribution is stationary. By the standard properties of a Markov process, it follows that it is the equilibrium distribution, so that the long term probabilities of being in each of the three states are known.

Solution to question 3:

The transition matrix in this case has the following properties:

$$\sum_{\text{all } i} p_{ij} = 1 \quad \text{and} \quad \sum_{\text{all } j} p_{ij} = 1.$$

The distribution that gives equal probabilities to all states is $\underline{\pi} = \frac{1}{N}\underline{e}^t$ where $\underline{e}$ is a $(n \times 1)$ vector with all elements being 1. So,

$$\begin{aligned}\underline{\pi}P &= \frac{1}{N}\underline{e}^tP = \frac{1}{N} \begin{pmatrix} 1 & 1 & \dots & 1 \end{pmatrix} P \\ &= \frac{1}{N} \begin{pmatrix} \sum_{\text{all } i} p_{i1} & \sum_{\text{all } i} p_{i2} & \dots & \sum_{\text{all } i} p_{iN} \end{pmatrix} \\ &= \frac{1}{N} \begin{pmatrix} 1 & 1 & \dots & 1 \end{pmatrix} = \underline{\pi}\end{aligned}$$

Thus this is the equilibrium distribution.

Solution to question 4:

Using Bayes formula, we have

$$\begin{aligned}P(N(s) = k | N(t) = n) &= \frac{P(N(t) = n | N(s) = k) P(N(s) = k)}{P(N(t) = n)} \\ &= \frac{P(N(t-s) = n-k) P(N(s) = k)}{P(N(t) = n)} \\ &= \frac{e^{-\lambda(t-s)} (\lambda(t-s))^{n-k} e^{-\lambda s} (\lambda s)^k}{e^{-\lambda t} (\lambda t)^n}\end{aligned}$$

$$(\text{which on simplification leads to}) = \binom{n}{k} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k} = \binom{n}{k} p^k (1-p)^{n-k}$$

where $p = \frac{s}{t}$.

Solution to question 5:

Let $X(t) = N(t) + M(t)$. Now,

$$\begin{aligned} P(X(t+dt) = h+dt | X(t) = h) &= P(\text{arrival in } N) + P(\text{arrival in } M) - P(\text{arrival in both}) \\ &= \lambda dt + o(dt) + \mu dt + o(dt) - o(dt) \\ &= (\lambda + \mu) dt + o(dt) \end{aligned}$$

As usual, any more than one event at one given time constitutes a probability of $o(dt)$. Thus

$$P(X(t+dt) = n+k | X(t) = k) = \begin{cases} (\lambda + \mu), & k = 1; \\ 1 - (\lambda + \mu)dt + o(dt), & k = 0; \\ o(dt), & k > 1. \end{cases}$$

which by definition is a Poisson process with rate $(\lambda + \mu)$.